A denominator bound for a contiguous basic hypergeometric ratio

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Abstract

We prove Kim and Stanton's Conjecture 8.3 in the final 2023 version of *Three families of q-Lommel polynomials*. A quadratic coefficient recurrence, obtained directly from the basic hypergeometric equation, preserves the conjectured denominator bound. The argument is over an integer polynomial ring and applies to every coefficient.

Task 1: the conjecture and its resolution

Let a, b, c, q be indeterminates, put $K = \mathbb{Q}(a, b, c, q)$, and work in $K[[z]]$. Write $(x; q)_m = \prod_{j=0}^{m-1}(1 - xq^j)$, with empty product 1, and set

$$F(z) = {}_2\phi_1(a, b; c; q, z) = \sum_{m\geq 0} \frac{(a; q)_m (b; q)_m}{(c; q)_m (q; q)_m} z^m, \qquad G(z) = {}_2\phi_1(aq, bq; cq; q, z).$$

Since $F(0) = 1$, its inverse exists. Define

$$R(z) = \frac{G(z)}{(1 - c)F(z)} = \sum_{n\geq 0} r_n z^n, \qquad D_n = \prod_{k=0}^{n} (1 - cq^k)^{\lfloor (n+1)/(k+1) \rfloor}.$$

Thus $r_n = \gamma_n(a, b, c)/(1 - c)$ in the notation of the conjecture.

Theorem 1. *For every $n \geq 0$, $P_n := D_n r_n$ belongs to $\mathbb{Z}[a, b, c, q]$. Consequently MathDB 350746, equivalently Conjecture 8.3 of Kim–Stanton [1], holds in full.*

The assertion is a denominator bound, not an assertion that D_n is the reduced denominator or that P_n has nonnegative coefficients. The latter is not part of the source conjecture.

Lemma 1. *Put $A = (1 - a)(1 - b)$. Then $r_0 = 1/(1 - c)$ and, for $n \geq 1$,*

$$(1 - cq^n) r_n = \big(a + b - ab - abq^n\big) r_{n-1} - Ac \sum_{i+j=n-1} q^j r_i r_j + Aabq \sum_{i+j=n-2} q^j r_i r_j. \tag{1}$$

Every summation index is nonnegative; a sum with negative total is zero.

Proof. If $f_m = [z^m]F(z)$, the defining product gives, for $m \geq 1$,

$$(1 - q^m)(1 - cq^{m-1}) f_m = (1 - aq^{m-1})(1 - bq^{m-1}) f_{m-1}.$$

Multiplying this equality by qz^m and summing, or simply comparing coefficients, yields

$$(c - abqz)F(q^2z) - \big(c + q - (a+b)qz\big)F(qz) + q(1-z)F(z) = 0. \tag{2}$$

The same product formula also gives

$$F(z) - F(qz) = \frac{Az}{1-c}G(z). \tag{3}$$

In fact, the coefficient of z^{m+1} on the left is $(1-q^{m+1})f_{m+1} = A[z^m]G(z)/(1-c)$.

Let $H(z) = F(qz)/F(z)$. Equation (3) says $H(z) = 1 - AzR(z)$, and $F(q^2z)/F(z) = H(qz)H(z)$. Divide (2) by $F(z)$, substitute these expressions, and cancel the common factor Aqz. This is legitimate over $K[[z]]$, where Aq is nonzero and multiplication by z is injective. The result is

$$\begin{aligned} R(z) - cR(qz) = 1 + z\big((a+b-ab)R(z) - abqR(qz)\big) \\ - AczR(z)R(qz) + Aabqz^2R(z)R(qz). \end{aligned} \tag{4}$$

The constant coefficient gives $r_0 = 1/(1-c)$. Taking the coefficient of z^n for $n \geq 1$ gives (1). □

Lemma 2. *For $n \geq 1$, put $E_n = D_n/(1-cq^n)$. In $\mathbb{Z}[c,q]$ one has*

$$D_{n-1} \mid E_n, \qquad D_iD_j \mid E_n \quad \text{if } i,j \geq 0 \text{ and } i+j \leq n-1.$$

Proof. The factor with $k = n$ in D_n has exponent 1, so $E_n = \prod_{k=0}^{n-1}(1-cq^k)^{\lfloor (n+1)/(k+1)\rfloor}$. For every $0 \leq k < n$,

$$\left\lfloor \frac{n}{k+1} \right\rfloor \leq \left\lfloor \frac{n+1}{k+1} \right\rfloor,$$

which proves the first divisibility. The exponent of the displayed factor in the product presentation of D_iD_j is

$$\left\lfloor \frac{i+1}{k+1} \right\rfloor + \left\lfloor \frac{j+1}{k+1} \right\rfloor \leq \left\lfloor \frac{i+j+2}{k+1} \right\rfloor \leq \left\lfloor \frac{n+1}{k+1} \right\rfloor.$$

Here a floor is zero when k exceeds the relevant index. As $i,j < n$, neither product has a factor indexed by $k \geq n$. Thus each quotient is explicitly a product of factors $1-cq^k$ with nonnegative integer exponents. No coprimality assertion is needed. □

Proof of Theorem 1. We have $D_0r_0 = 1$. Suppose $P_i = D_ir_i \in \mathbb{Z}[a,b,c,q]$ for all $i < n$. Multiplication of (1) by E_n gives the constructive recurrence

$$\begin{aligned} P_n = (a+b-ab-abq^n)\frac{E_n}{D_{n-1}}P_{n-1} \\ - Ac\sum_{i+j=n-1} q^j \frac{E_n}{D_iD_j}P_iP_j \\ + Aabq\sum_{i+j=n-2} q^j \frac{E_n}{D_iD_j}P_iP_j. \end{aligned} \tag{5}$$

Every quotient in (5) lies in $\mathbb{Z}[c,q]$ by Lemma 2. All remaining factors are integer polynomials by induction. This proves $P_n \in \mathbb{Z}[a,b,c,q]$ and completes the induction. □

For example,

$$P_0 = 1, \qquad P_1 = a + b - ab(1 + q) - c + abcq.$$

The proof is a formal rational-function identity. It therefore covers all specializations where the coefficients are defined, including $a = 1$ or $b = 1$; the temporary generic cancellation of A loses no such cases. At poles, the assertion retains its rational-function meaning. Values such as $q = 1$, where the original coefficient formula is singular, may only be interpreted after any indicated rational continuation.

Relation to earlier work. The recurrence method extends the q-Kishore strategy used by Kim and Stanton [2], especially their Proposition 3.3 and Theorem 3.2 for Jackson's first q-Bessel function. Their earlier manuscript states the present full-parameter assertion as Conjecture 7.2; the final paper [1] retains it as Conjecture 8.3. The introduction of the final paper attributes the earlier q-Kishore results to Lalanne's work of 1992–1993; those special cases are not new results here. The new step is (1) for arbitrary a, b, c, with the integral induction (5). All identities used in the proof have been derived above.

Proposed follow-up Tasks 2–5

These are research directions, not additional solved claims.

2. Determine, for each n and $0 \le k \le n$, the exact generic pole order of r_n along $1 - cq^k = 0$, and hence the reduced denominator.

3. Find a finite weighted-tree formula for P_n realizing all three terms of (5), with polynomial edge or vertex weights.

4. Classify the specializations $a = q^{-u}, b = q^{-v}$, for nonnegative integers u, v, for which the pole orders are smaller than their generic values, and give the resulting reduced denominator in $\mathbb{Q}(q)[c]$.

5. For fixed nonnegative integers u, v, w, determine a denominator bound in $\mathbb{Z}[a, b, c, q]$ for the coefficients of ${}_2\phi_1(aq^u, bq^v; cq^w; q, z)/{}_2\phi_1(a, b; c; q, z)$, including necessary factors that depend only on q.

References

[1] J. S. Kim and D. Stanton, *Three families of q-Lommel polynomials*, Séminaire Lotharingien de Combinatoire **87B** (2023), Article 8, 25 pp., Conjecture 8.3, p. 24. Final article.

[2] J. S. Kim and D. Stanton, *Ratios of Jackson's q-Bessel functions and q-Lommel polynomials*, author manuscript dated 7 September 2020, 25 pp., Sections 2–3 and Conjecture 7.2, pp. 23–24. Author manuscript.